Report on *Matrix Theory Reloaded: A BPS Road to Holography*

(Blair, Lahnsteiner, Obers, Yan; file `paper.tex`)

The paper revisits the BPS decoupling limits that give matrix theory on Dp-branes (MpT), and asks how they relate to holography. Three questions are posed in the introduction; the three sections below answer the three items requested in `AGENTS.md`.

1 Principal achievement

The central result is a geometric identification. Matrix theory arises from a BPS decoupling limit of the Dp-brane worldvolume theory, which in the flat background is the limit $\omega \to \infty$ of the metric $\omega\, \mathrm{d}x^A \mathrm{d}x^B \eta_{AB} + \omega^{-1} \mathrm{d}x^{A'} \mathrm{d}x^{A'}$. This produces a non-Lorentzian target space with a codimension-$(p+1)$ foliation. For the near-horizon $\mathrm{AdS}_5 \times S^5$ geometry, the bulk metric

$$\left(\tfrac{r}{\ell}\right)^2 \mathrm{d}x^A \mathrm{d}x^B \eta_{AB} + \left(\tfrac{\ell}{r}\right)^2 (\mathrm{d}r^2 + r^2 \mathrm{d}\Omega_5^2)$$

has exactly the M3T form with $\omega = (r/\ell)^2$. The near-horizon limit in the bulk and the asymptotic matrix-theory limit on the boundary are therefore the same BPS limit, which answers Question 2 precisely.

The paper then draws two consequences from this:

- The bulk geometry is an intrinsic deformation of the asymptotic non-Lorentzian geometry. Undoing the BPS limit gives a p-brane generalisation of $T\bar{T}$ (Question 3).
- This deformation gives flow equations, of the form $\partial\mathcal{L}/\partial\lambda$ equal to a function of $T_{\alpha\beta}$, that take free scalars to the Dirac–Born–Infeld action. The case $p = 1$ reproduces the standard $T\bar{T}$ flow, and $p = 2$ gives an explicit new three-dimensional flow (the paper's membrane $T\bar{T}$ equation in the introduction).

The DLCQ^n/DLCQ^m picture then organises the whole duality web of limits in a single framework. Taken together, this is a coherent and unusually broad framework that ties matrix theory, AdS/CFT, non-Lorentzian geometry and $T\bar{T}$ together.

Check on the $p = 1$ flow. The paper states that its general p-brane flow equation reduces, at $p = 1$ with $F = 0$, to the standard $T\bar{T}$ flow. I checked this numerically on the explicit free-boson (Nambu–Goto) solution, using a covariant density $\sqrt{-\gamma}$ and finite-difference derivatives. At $\lambda = 0.1$ and $\lambda = 0.3$ the three quantities $\partial_\lambda \mathcal{L}$, $\frac{1}{2} \det T_{\alpha\beta}$ and $-\operatorname{tr} \mathcal{T}/(2\lambda)$ agree to six digits. The claim holds on this solution.

2 Principal drawback

The new holographic claims are not yet tested dynamically. The paper establishes geometry and limits, but it does not check that the new backgrounds are solutions of a theory. Four points support this:

1. The non-Lorentzian bulk geometries beyond AdS (Sections on non-Lorentzian holography) are not shown to satisfy the equations of motion and torsional constraints of MpT supergravity. The Outlook says so explicitly, and MpT supergravity is not formulated in the paper.

2. The examples depend on an extra rescaling of the length ℓ in the harmonic function, which the authors introduce "by hand". They say its physical meaning is "not completely clear" and treat it as "a trick". It is also equivalent to rescaling N, which relates it to a large-N limit whose interpretation remains open.

3. The DLCQ^n/DLCQ^m correspondence, which the abstract presents as the paper's conjecture for holography, comes with no spectrum, correlator or symmetry test.

4. The p-brane $T\bar{T}$ flow equations are classical, bosonic and abelian. They are complete only for $p = 1, 2$. For $p = 3$ they are given only to linear order in the flow parameter, and for $p > 3$ only at zeroth order.

The first two points are the most serious. Without them, the flagship non-Lorentzian holographic examples remain kinematic (geometry plus limits) rather than verified backgrounds of a theory, and the paper's central new claim is untested.

3 Most worthwhile follow-up: feasibility versus impact

Recommendation: compute the isometry algebra of one simple non-Lorentzian bulk geometry from the paper, and compare it with the symmetry algebra of the boundary MpT theory.
The paper names this calculation itself in the Outlook: "compute the isometry group of this type of non-Lorentzian geometries."

- **Concrete steps.** Take the simplest explicit example in the paper, preferably the lowest p with an explicit geometry. Write down the Killing equations appropriate to the vielbein (torsional Newton–Cartan) structure and solve them with a computer algebra system such as SageMath or Cadabra. Compare the resulting Killing algebra with the p-brane boost and the Galilean-type symmetries of the boundary MpT theory.

- **Outcomes.** If the bulk symmetries match the boundary, this supports the DLCQ^n/DLCQ^{n+1} picture at the level of symmetries. If they do not, the proposed duals are in trouble or need modification, and either result is publishable.

- **Feasibility: high.** The metrics are explicit, the method is standard, and the work needs no new formalism beyond the vielbein Killing condition, so it is a modest, self-contained project.

- **Impact: high.** It is the most direct first test of the paper's central new claim, and it gives the dynamical check that the paper currently lacks. It is also a prerequisite for the supergravity equation-of-motion check, which is the natural next step but needs MpT supergravity to be formulated first.

By comparison, a non-abelian p-brane $T\bar{T}$ flow (which would deform matrix theory to DBI) is arguably higher impact but has low feasibility and is open-ended. Extending the construction to heterotic strings is a longer-term programme, as the paper's Outlook also suggests.